

COMP9319 Exercises

Answers : To be released one week later.

Question 1

Suppose that the BWT encoded string $BWT(T) = \text{arbrraa\$}$

where $\$$ is the last character of T .

Derive the number of matches for the search pattern **ar** using backward search.

Question 2

Suppose that the BWT encoded string $BWT(T) = \text{acb\$cccbaabbcab}$

where $\$$ is the last character of T .

Derive the number of matches for the search pattern **abc** using backward search.

Question 3

Suppose that the BWT encoded string $BWT(T) = \text{n\$rsoocimpse}$

Derive the S , B , and B' arrays after applying RLFM index on T .

Question 4

Suppose that the RLFM encoded string of text T is **cgcs\$agagatc** where $\$$ is the last character of T . Its corresponding bit array B is **1101011101110011**.

Derive its B' .

Question 5

Suppose that the RLFM encoded string of text T is **cgcs\$agagatc** where $\$$ is the last character of T . Its corresponding bit array B is **1101011101110011**.

Derive the number of matches for the search pattern **cag** using backward search.

Question 6

Suppose that the RLFM encoded string of text T is **cgcs\$agagatc** where $\$$ is the last character of T . Its corresponding bit array B is **1101011101110011**.

Derive the last 4 characters of T .